Vanshika Sharma

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SUMMARY

Third-year BTech Computer Science & Engineering (**Quantum Computing Honors**) student at **SRM Institute of Science & Technology** with an exceptional academic record (**CGPA: 9.8**), driven by a strong **research orientation** and hands-on engineering experience across **Artificial Intelligence, Full-Stack Systems, Cybersecurity, and Post-Quantum Cryptography**.
Former **Research Intern at IIT Kanpur and IIT Ropar**, with focused contributions in **Hardware Roots of Trust, secure boot architectures, quantum-resilient cryptographic systems, and real-time distributed platforms**. Demonstrated ability to translate **theoretical security and cryptographic models into scalable, production-ready system designs**.
Winner of multiple **national-level hackathons**, with proven experience building **high-impact AI products** including **real-time meeting intelligence platforms, biometric authentication systems, quantum-enhanced phishing detection pipelines, and neuroscience-driven analytics dashboards**. Strong track record of **rapid prototyping, system optimization, and end-to-end product ownership**.
Technically proficient in **C, C++, Python, Java, React, Node.js, MongoDB, Supabase, TensorFlow, system architecture, and cryptographic engineering**, complemented by strong **problem-solving ability, research execution discipline, leadership, and cross-functional collaboration**.

EDUCATION

Bachelor of Technology - Computer Science & Engineering 2023 – 2027
SRM Institute of Science and Technology Chennai, India
CGPA 9.8 Relevant Coursework- DSA, Operating Systems, OOP, Advanced Programming Practices, Computer Organization & Architecture, Cryptography.

Class 12th : The Millennium School 2023
Percentage: 89% (PCM)

Class 10th : The Millennium School 2021
Percentage: 92.4%

PROFESSIONAL EXPERIENCE

Research Intern: IIT Ropar 🔗 May 2025 – Jul 2025
Dopamine Dashboard (DDD): Neuro-Analytics Platform
Tech Stack: MERN (MongoDB, Express.js, React.js, Node.js), REST APIs

- Architected and developed a **real-time neuro-feedback analytics dashboard** to monitor **dopamine-level variations and associated cognitive behavior patterns** using the **MERN stack**.
- Built **interactive data visualizations and dynamic dashboards** to improve **neuroscience data interpretability** for both technical researchers and non-technical stakeholders.
- Designed and integrated **RESTful APIs with asynchronous data pipelines** for **low latency data updates and seamless sensor to dashboard communication**.
- Implemented **real-time state management and optimized backend performance**, ensuring scalability and reliable data synchronization.
- Collaborated with **neuroscientists and research teams** to map **behavioral metrics with neurochemical variations**, enabling accurate **behavior to biology correlation models**.
- Applied principles of **system design, real-time analytics, and human centered UI/UX** for research-grade data platforms.

IIT Kanpur: Research Intern 🔗 Jul 2025
Department of Computer Science & Engineering Mentor: Prof. Angshuman Karmakar **Project:** *Role of Hardware Root of Trust in the Post-Quantum Era*

- Conducted advanced research on **Hardware Root of Trust (RoT) architectures** for securing **classical–quantum hybrid systems** against **post-quantum adversarial threats**.
- Analyzed **security vulnerabilities in cryptographic hardware modules** (TPMs, secure elements, HSMs) and proposed **post-quantum–compatible trust anchors** to strengthen **system integrity and secure boot chains**.
- Evaluated **quantum-safe cryptographic primitives**, including **lattice-based and hash-based schemes**, for deployment within **secure hardware enclaves**.
- Designed **theoretical models for quantum-resistant trust chains**, enabling **secure boot mechanisms and firmware authenticity validation** in **quantum-capable embedded systems**.
- Performed **literature-driven benchmarking of real-world RoT deployments** across **IoT, financial infrastructure, and critical national systems**.

Media and Outreach Head 2024 – present
IEEE CS SRM 🔗

- Led marketing campaigns for **5+ technical events**, increasing event participation by **30%** through targeted outreach.
- Successfully drove community engagement via social media and email campaigns, leading to a **15% growth** in the IEEE CS SRM network.

Quantum Core Lead and Web Associate 2023 – present
Quantum Computing Club SRM

- Serve as **Quantum Core Lead**, driving **technical initiatives, sessions, and mentorship** in quantum computing and post-quantum security.
- Managed **digital platforms and member engagement**, achieving a **20% increase in active participation**.
- Led the organization of **Quantum 2.0 Hackathon** with **75+ teams**, enabling large-scale innovation and collaboration.

CERTIFICATES

- Python for Data Science 🔗
- Azure Fundamentals Cloud Skill Challenge 🔗
- Github Foundation 🔗
- AI-ML Virtual Internship 🔗
- TEDx 🔗
- Salesforce Certifed Agentforce Specialist 🔗
- Android Developer Virtual Internship 🔗
- e-Yantra Course 🔗
- Oracle 🔗
- IEEE 🔗
- Microsoft 🔗

RESEARCH PUBLICATION & CONFERENCE PRESENTATION

Hardware Anchors for Post-Quantum Security : ICEdge 2025 (IEEE, IISc Bangalore)
Co authored and presented a research paper introducing a Quantum Root of Trust (Q-RoT) architecture that combines Post-Quantum Cryptography (PQC), Quantum Random Number Generators (QRNG), and Physically Unclonable Functions (PUFs) to establish quantum-resilient secure boot, trusted device identity, and hardware-anchored authentication for future-ready edge, IoT, and critical infrastructure systems.

Integrating Quantum Computing with Deep Learning for Early Disease Detection — ICDMAI 2026
Proposed a hybrid Quantum AI architecture combining parameterized quantum circuits with deep neural networks to enhance early disease prediction, optimize feature learning, and improve diagnostic reliability across healthcare datasets, validating performance gains in accuracy, precision, recall, and robustness.

AWARDS

Winner - Flex-It-Out Hackathon — Feb 2025
Bajaj Finserv Health
- Secured **1st place among 300+ teams** in the Flex-It-Out Hackathon.
- Developed an **AI-powered pose detection system for fitness**, integrating a **yoga detection module** for enhanced accuracy and real-time feedback.
- **Won a cash prize of 25k** for the innovative solution, recognized for its impact on health and fitness.
- Leveraged **AI/ML techniques** to improve health outcomes and collaborated with a **diverse team** to design and implement the winning project

Academic Achievement in Programming — Nov 2024
Recognition in Java
- Ranked **top 5% in NPTEL Programming in Java** among **25,000 participants**, showcasing strong expertise in Java development.
- Earned the **Silver Zone award**, highlighting excellence in problem-solving.

Pre-Finalist – Unisys Innovation Program (Year 16)
Project: *Quantum Cybersecurity for Phishing Detection using QRNG*
- Selected as a **Pre-Finalist** for developing a **quantum-enhanced phishing detection system** using **Quantum Random Number Generation (QRNG)**.
- Designed a **post-quantum secure cybersecurity framework** to counter evolving **phishing and social engineering attacks**.
- Applied **quantum entropy, cryptographic randomness, and secure system design principles** for next-generation threat detection.
- Recognized by **Unisys Global Innovation Team** for **scalability, security architecture, and applied quantum research potential**.

Winner: IntelliHack National AI Hackathon — Apr 2025
Organizers: MeetStream AI, Nyx Wolves & School of Computing, SRMIST
- Secured **1st Place** at a **national-level AI hackathon** for building **MeetSight**, an **AI-powered real-time meeting intelligence platform**.
- Developed an **agentic AI system** that performs **live transcription, task detection, Jira automation, real-time data visualization, and contextual memory search**.
- Integrated **LLMs (Gemini 1.5, GPT-4o), React, Tailwind, Slack, Jira, Notion, Google Sheets, Salesforce, and Miro**.
- Demonstrated **end-to-end full-stack AI system design, automation workflows, and real-time decision support systems**.
- Awarded **$5,000 Jigsaw Stack credits + 900 Superflex AI credits** for innovation excellence.

Winner Evincible Ideathon — Nov 2023
SRMIST
- Secured the **1st position** among **30+ teams** with "Green Ledger," a blockchain-based solution for tracking carbon credits.
- Delivered compelling presentations, demonstrating strong **communication and presentation skills** to a jury panel

Project Expo — Mar 2025
organized by CTech dept SRMIST
- Ranked in the **top 10 out of 100 teams** at the **Project Expo of the CTech Department** for an **agro-based IoT and software solution**.
- Developed **Farm and Supply**, integrating IoT and software technologies to enhance agricultural supply chain efficiency.

SKILLS

Languages — C, C++, Python, Java, OOPs

Web Development — HTML, CSS, JavaScript, ReactJS, Next JS, Tailwind CSS

Other tools — Figma, VS Code, Github, Linux ,UI, Machine Leaning,AI

Soft Skills — Problem-Solving, Teamwork, Communication, Leadership, Technical Discussions, Networking

PROJECTS

QCyber
QCyber a quantum-enhanced phishing detection system where I contributed.
This project explores the use of quantum principles to improve the robustness of phishing detection models, inspired by the future need for post-quantum secure systems.

Faunalyze – Fauna + Analyze (intelligent wildlife analysis)
- Developed an advanced **image-based animal species classification system** using transfer learning with the **VGG16 model** and **Convolutional Neural Networks (CNNs)**.
- Trained the system with a dataset of **10,00+ labeled images** across six animal species to achieve **85% accuracy** in classification.
- Aimed to aid conservationists in identifying wildlife and discovering new habitats and species.